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LAB EXERCISE2 – Logistic

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```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn import metrics
import seaborn as sns
```

```
url = "https://stats.idre.ucla.edu/stat/data/binary.csv"

df = pd.read_csv(url)

df.head()
```

	admit	gre	gpa	rank
0	0	380	3.61	3
1	1	660	3.67	3
2	1	800	4.00	1
3	1	640	3.19	4
4	0	520	2.93	4

```
print("Number of rows:", df.shape[0])

print("Number of columns:", df.shape[1])

print("\nColumn names:")

print(df.columns.tolist())
```

Number of rows: 400
Number of columns: 4

Column names:
['admit', 'gre', 'gpa', 'rank']

```
df.isnull().sum()
```

```

      0
admit 0
gre    0
gpa    0
rank   0

dtype: int64
```

```
df.head()
```

	admit	gre	gpa	rank
0	0	380	3.61	3
1	1	660	3.67	3
2	1	800	4.00	1
3	1	640	3.19	4
4	0	520	2.93	4

```
feature_cols = ["gre", "gpa", "rank"]

x = df[feature_cols]

y = df["admit"]
```

```
print(x)

print(y)
```

```

      gre  gpa  rank
0    380  3.61     3
1    660  3.67     3
2    800  4.00     1
```

```

3    640  3.19    4
4    520  2.93    4
..    ...    ...    ...
395  620  4.00    2
396  560  3.04    3
397  460  2.63    2
398  700  3.65    2
399  600  3.89    3

```

```
[400 rows x 3 columns]
```

```

0    0
1    1
2    1
3    1
4    0

```

```

..
395    0
396    0
397    0
398    0
399    0

```

```
Name: admit, Length: 400, dtype: int64
```

```

x_train, x_test, y_train, y_test = train_test_split(
    x,
    y,
    test_size=0.25,
    random_state=0
)

```

```

print(x_train.shape)
print(y_train.shape)
print(x_test.shape)
print(y_test.shape)

```

```

(300, 3)
(300,)
(100, 3)
(100,)

```

```

model = LogisticRegression(
    solver='lbfgs',
    max_iter=1000
)

```

```
model.fit(x_train, y_train)
```

```

▼ LogisticRegression ⓘ ?
LogisticRegression(max_iter=1000)

```

```

y_pred = model.predict(x_test)

print(y_pred)

```

```

[0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 0 0 1 0 0 1 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 1 0 0 1 0 0 0 0 1 0 0 0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0]

```

```
print(y_test.values)
```

```

[0 0 1 0 0 1 1 0 1 1 0 0 0 0 1 0 0 0 0 0 0 1 0 0 1 1 0 0 1 0 0 1 1 1 0 0 0
 1 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 1 1 0 1 0 1 0 1 0 0 0 0 0 0 0
 0 1 0 0 1 0 0 0 1 1 0 1 0 1 0 0 0 1 0 1 1 0 0 1 1 1]

```

```

conf_mat = metrics.confusion_matrix(
    y_test,
    y_pred
)

print(conf_mat)

```

```

[[61  4]
 [27  8]]

```

```

accuracy = metrics.accuracy_score(
    y_test,
    y_pred
)

```

```
print("Accuracy:", accuracy)
```

```
Accuracy: 0.69
```

```
print("Accuracy in percentage:", accuracy * 100, "%")
```

```
Accuracy in percentage: 69.0 %
```

```
conf_mat = pd.crosstab(  
    y_test,  
    y_pred,  
    rownames=["Actual"],  
    colnames=["Predicted"]  
)  
  
sns.heatmap(  
    conf_mat,  
    annot=True  
)
```

<Axes: xlabel='Predicted', ylabel='Actual'>

